

white-eyed fly was male. This showed that the white-eyed trait must be linked to the fly's sex. Further experiments revealed other sex-linked physical traits, and Morgan concluded that all these traits must be inherited together from the chromosome responsible for sex determination. Male sex chromosomes are X and Y, whereas the female sex chromosomes are both X, so Morgan could deduce that white eyes and other traits seen only in males must be carried

on the Y sex chromosome. Further study revealed that genes occupy a specific location on a chromosome, and this enabled the production of genetic maps.

Through his fly experiments, Morgan confirmed the chromosomal theory of inheritance, and acknowledged the value and accuracy of Mendel's work. In extending Mendel's work on plants and applying it to animals, Morgan marked a turning point in the study of inheritance and launched the field of genetics.

STUDIED FRUIT FLIES FOR 17 YEARS



WROTE 370 SCIENTIFIC PAPERS

